IKJYOT SINGH

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EDUCATION

University of Alberta - Bachelor of Science in Computing Science with Specialization - GPA 3.5 August 2022 – April 2026

Relevant Coursework: OOP, Algorithms, Data Structures, Search & Planning in AI, Reinforcement Learning, Computer Vision, Cybersecurity

EMPLOYMENT EXPERIENCE

AI Developer – So Shall We Inc. December 2024 – September 2025

- Engineered and deployed a RAG-enabled customer support chatbot on AWS (Lambda, DynamoDB) with automated ETL pipelines for monthly refreshes and continuous low-latency inference.
- Decoupled recommendation logic from the LLM by developing a recommendation engine using TF-IDF, cosine similarity, and co-occurrence matrices, reducing GPT token costs by ~20% and improving response latency.
- Containerized ML services with Docker to bypass AWS Lambda's strict package size limits, maintaining a lightweight footprint.

Software Developer – Dr. Dash's Lab, Faculty of Rehabilitation Medicine, University of Alberta May 2024 – August 2025

- Assisted in NLP analysis of verbal fluency data from 127 patients, using cosine similarity to measure semantic distance and identify differences in semantic mapping between Alzheimer's patients and healthy controls.
- Developed a Python program to parse 242 .cha files and clean ~8,000 datapoints using Pandas.
- Researched word embedding, semantic granularity, and the extraction of lexicosemantic features from patients' words.

Computer Vision Research Assistant – Dr. Cummine's Lab, Faculty of Rehabilitation Medicine, U of A May 2023 – December 2023

- Developed a CNN model to classify patients with learning impairments using TensorFlow using Multi-Instance Learning.
- Automated the pre-processing (parsing and cropping) of 8,000+ 3-dimensional MRI images.
- Implemented methods to counteract overfitting including data augmentation, k-fold cross validation, LOO-CV, and L2 regularization. Implemented feature analysis methods such as SHAP, Grad-CAM, Saliency maps, and LIME.

Senior Residence Assistant – University of Alberta, Tamarack Residence August 2025 – May 2026

- Led a collaborative team of 9 Resident Assistants (RAs) by encouraging accountability, support, and communication.
- Acted as a point of escalation for RAs, supporting decision-making in complex or time-sensitive situations.
- Managed high-pressure situations involving resident safety, conflict, and well-being.

PROJECTS

Marvel Movie Bot: Context-Rich Chatbot Using Retrieval-Augmented Generation. June 2025

- Built a RAG-powered chatbot using a local LLaMA model and Pinecone vector store of 19 Marvel movie scripts.
- Developed a React frontend, fastAPI backend, and exposed endpoints with ngrok; deployed frontend on Vercel.
- Tuned chunking and retrieval to outperform standard LLaMA responses in detail and contextual accuracy.

Soccer Analysis System: Tracking Game and Calculating Relevant Statistics August 2024

- Trained YOLO model with open-source data to track players and ball and various possession statistics in a soccer game.
- Used OpenCV feature analysis to keep track of camera movement. Created custom interface outputted as AVI video.
- Estimated speed and distance statistics incorporating Hash Tables and Linked Lists.

Fly Chicken Fly!: Endless runner and platformer video game with unique game mechanics. May 2024

- Leveraged C# and Unity to develop movement mechanics and structure for seamless music & sound effect integration.
- Utilized Unity's animation and physics systems to refine gameplay and optimize player feedback.
- Composed and implemented soundtracks using Audacity; Nominated for Best Sound Design Award.

PUBLICATIONS

Patel, D., Rizk, R. & **Singh, I.** (2026). Developing a Diagnotic Tool for Assessing Missingness Mechanisms in Healthcare Datasets. STEM Fellowship Journal (National Inter-University Big Data Challenge). [DOI]

SKILLS

- **Languages:** Python, Java, JavaScript, C#, C, R, SQL, HTML, CSS
- **Tools/Frameworks:** Git, JUnit, Google Cloud Platform (Firestore), React.js, Node.js, TensorFlow, PyTorch, REST APIs, RAG Pipelines, AWS Lambda & DynamoDB, Cursor, Docker
- **Libraries:** Pandas, NumPy, Matplotlib, PyDicom, OpenCV